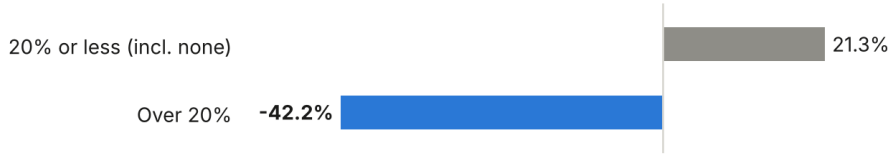
Where the superstore loses money: heavy discounts

Question: which products, regions or customer segments lose money, and what should the business change?

> **Sales discounted by more than 20% earned a margin of -42.21%, against 21.30% for sales discounted by 20% or less (including none), from 2011 to 2014.**

Sales discounted over 20% earned a margin of -42.21%

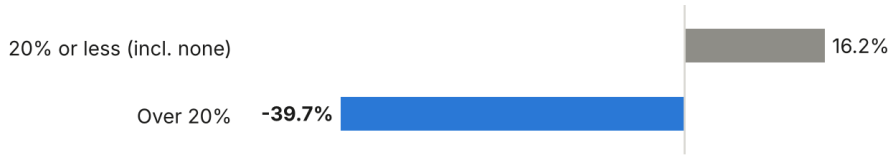
Profit margin by discount size, 2011 to 2014. 11,328 of 51,290 order lines are over 20%.



Source: Global Superstore (Kaggle, fatihilhan), 51,290 order lines. Margin = total profit / total sales.

Tables lines discounted over 20% earned a margin of -39.68%

Tables only, 2011 to 2014. 467 of Tables' 861 order lines are over 20%.



Source: Global Superstore (Kaggle, fatihilhan), 51,290 order lines. Margin = total profit / total sales.

What the numbers show

Data: 51,290 order lines (one product on one order) from 2011 to 2014. Sales were 12,642,905 and profit 1,467,457.29 in currency units, a margin of 11.61%. Margin means total profit divided by total sales. A heavy discount means over 20%.

- **Heavy discounts.** 11,328 order lines carried one. 89.9% of them lost money, against 5.9% of the lines discounted 20% or less. The pattern holds in every category, customer segment and year, and in all 12 regions that have any heavily discounted lines. Among the 5,369 products sold both at 20% or less and over 20%, their lines earn 21.46% at 20% or less and -42.31% above 20%.
- **Tables** (a furniture sub-category) is the only sub-category with a net loss, and it lost money in all four years: -64,083.39 on sales of 757,034, a margin of -8.47%. Its 467 lines discounted over 20% have a net loss of 132,541.15, the largest of any sub-category. Its 208 lines with no discount made 60,447.20 and none of them lost money.
- **Regions and customer segments.** No region or segment loses money overall. The three segments sit within half a point of each other (11.51% to 11.99%). The weakest regions are Southeast Asia (2.02%) and EMEA (5.45%).
- **Two other explanations did not hold up.** Shipping cost as a share of the sale did not push margin down: it stayed between 9% and 19% in every group of 40 or more lines. And no category loses money, with Technology the highest at 13.99%.

Recommendation

I'd test a 20% discount cap on Tables first. It holds the largest heavy-discount loss of any sub-category: 467 of its 861 order lines are discounted over 20%, with a net loss of 132,541.15. Run the cap for two quarters and compare Tables' margin with the other sub-categories over the same period. It's -8.47% now, and its lines at 20% or less already earn 16.18%, so a rise toward that would mean the cap is working.

Limits

- **Volume is untested.** The data cannot show whether a cap would cost sales. As a rough bound, if every over-20% Tables line disappeared, Tables profit would move from -64,083.39 to +68,457.76. What the data cannot show is knock-on effects, such as customers who leave and take other purchases with them.
- **The 20% cutoff was partly picked after I looked at profit by discount level.** Margin falls steadily as discounts grow: 10.05% at 15% to 20%, 0.20% at 20% to 25% (239 lines), -7.68% at 25% to 30%. Treat 20% as a cautious cap, not a precise threshold.
- **This is an association, not proof of cause.** Comparing the same products rules out "just bad products", but I cannot rule out other reasons, such as why a discount was given.
- **The shipping result is weaker than the others.** The last two shipping groups hold only 14 and 3 sales, so I ignore them. Profit also appears not to include shipping cost, so this check is not conclusive.
- **The data is public Kaggle data and may be synthetic.** The currency is not stated, so figures are in currency units.